

Talent.exe

Skill Hub

Skills Intelligence Platform · AI-Powered Talent Discovery

Team : ./claude & chai

TalentExe — Platform Overview

Skills Intelligence Platform · Roles, stack, and where AI fits

USER ROLES

HR Team

- Manages the skills database & employee roster
- Reviews and approves AI-extracted profiles
- Searches talent using natural language (NLP)
- Downloads employee profile as PDF

Employee

- Uploads resume (PDF/DOCX) for AI extraction
- Reviews and corrects AI-extracted data
- Refines skills, projects & certifications
- Submits profile for HR approval

SYSTEM LAYERS

Next.js Frontend

App Router · Tailwind CSS · shadcn/ui · TypeScript

Django + DRF Backend

REST API · Supabase JWT auth · Role-based routing

Supabase (PostgreSQL + pgvector)

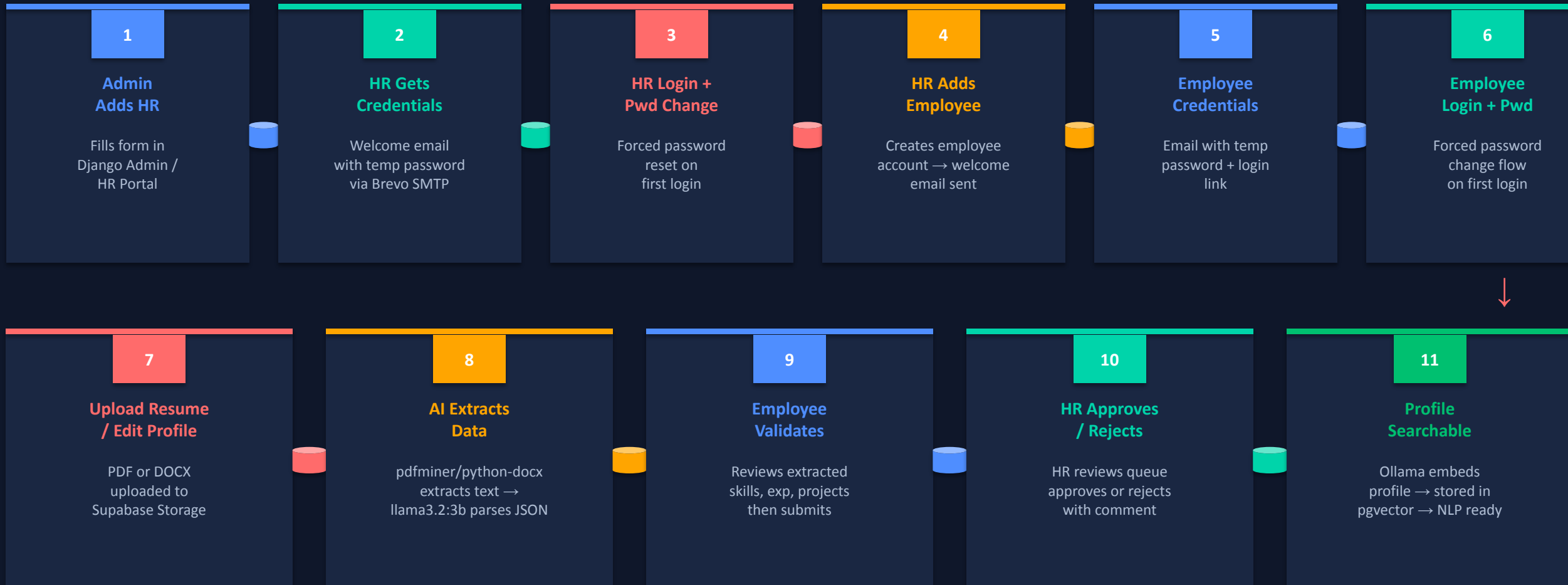
Profiles · Reviews · Resume storage · RLS policies · Vector store

AI Layer — Extraction & Search

Ollama llama3.2:3b → Entity extraction + NLP scoring
Ollama nomic-embed-text → 768-dim vector embeddings
pgvector → Cosine similarity search

Platform Flow — End-to-End Journey

From Admin adding HR to a profile becoming searchable by NLP



PDF Extract & AI Use — Ingestion Pipeline

Resume upload → text extraction → llama3.2:3b entity parsing → review queue → profile active

INGESTION PIPELINE

PDF/DOCX Upload

Employee uploads
resume ≤ 10 MB
.pdf / .doc / .docx
Stored in Supabase
Storage bucket

Text Extraction

pdfminer.six
for PDF files
python-docx for
DOCX files
Raw plain text out

Llama Extraction

llama3.2:3b
reads raw text →
outputs structured
JSON with all
entities + confidence

Skill Inference

llama3.2:3b also infers
implied skills
e.g. Next.js → React
Confidence scored
0.0 – 1.0 per item

Review Queue

Extracted data
held for HR &
employee review
before it enters
normalized tables

Profile Activated

On approval →
Ollama generates
768-dim embedding
→ pgvector store
→ NLP searchable

WHAT CLAUDE EXTRACTS FROM RESUME TEXT

Skills

name · proficiency
(Beginner/Intermediate
/Expert) · years

Experience

company · role · start
& end dates · job
description

Projects

name · tech stack
description
· duration

Education

degree · institution
year of graduation
· field of study

Certifications

cert name · issuer
issue date
· expiry date

PDF Extract & AI Use — Review & Confidence

Human-in-the-loop approval, profile states, llama3.2:3b vs Ollama embed responsibilities

LLAMA3.2:3b vs OLLAMA EMBED — CLEAR RESPONSIBILITIES

Ollama llama3.2:3b (LLM)

- Reads raw resume text → outputs structured JSON
- Extracts: skills, experience, projects, education, certs
- Assigns confidence score (0.0–1.0) to each extracted item
- Infers implied skills (e.g. 4 yrs Next.js → React expertise)
- Scores search results 0–100 against HR's NLP query
- Generates plain-English match explanation per result

Ollama (nomic-embed-text 768d)

- Converts employee profile text into a 768-dim vector
- Converts HR search query into a 768-dim vector
- Runs locally — no API key, no cost
- Model: nomic-embed-text (free, open-source)
- Vectors stored in pgvector for cosine similarity search

PROFILE STATE MACHINE

INCOMPLETE

Profile created — no data submitted yet



SUBMITTED

Employee validated AI data + hit Submit



APPROVED

HR approved → Ollama embeds → NLP searchable



REJECTED

HR rejected with comment → employee must re-edit

AI CONFIDENCE THRESHOLDS

Extraction threshold

≥ 0.7

Items below flagged for manual review

Inference threshold

≥ 0.6

Inferred skills below this are dropped

NLP Use & Flow — Search Pipeline

How a plain-English HR query becomes ranked, explained, scored results

6-STEP NLP SEARCH PIPELINE



TECHNOLOGY USED

Query Parsing & Scoring

Ollama llama3.2:3b

Understands ambiguous HR language · scores profiles · explains match

Embedding Generation

Ollama nomic-embed-text (768d)

Runs locally · free · no API key · same model for queries & profiles

Vector Similarity

pgvector cosine ($\leq$) operator

Enabled in Supabase · fast · finds semantically similar profiles

NLP Use & Flow — Scoring & UI

How llama3.2:3b calculates match scores, what HR sees, and the API contract

HOW NLP GIVES A SCORE — STEP BY STEP

1. Cosine Similarity

pgvector computes cosine angle between query vector and profile vector → raw similarity score 0.0 – 1.0

2. Hard Filter Pass

Profiles failing location / min-years / department filters are removed — they cannot receive a high score

3. llama3.2:3b Scoring Prompt

llama3.2:3b receives the top 20 profiles + original HR query. It is asked: score 0–100 for fit. It weighs skill matches, years of experience, seniority level, location & availability.

4. Score Bands

90–100 : all required skills + location matched → green
70–89 : most skills matched, minor gaps → amber
50–69 : partial match, some relevant exp → orange
< 50 : weak match, shown last, visually muted → grey

5. Plain-English Explanation

llama3.2:3b generates a 1–2 sentence human-readable reason: "Expert Java dev (6 yrs), Razorpay & PayU integration on 2 production projects. Based in Pune. Currently unallocated."

HR SMART SEARCH UI



"Find backend dev in Pune, 3+ yrs Java" |

[Search] |

Filter chips (auto-generated by llama3.2:3b parse):

[Skills: Java, payment gateway] [Location: Pune]

[Min Exp: 3 yrs] [Role: backend]

94% match ↗

Rahul Sharma · Sr. Backend Dev · Pune |
Expert Java (6 yrs), Razorpay & PayU, 2 prod |
projects. Currently unallocated. |
[Java] [Spring Boot] [Razorpay] [View Profile] |

API — POST /api/v1/search/

Request:

```
{ "query": "backend dev Pune 3+ yrs Java payment gateway" }
```

Response:

```
{ "query_parsed": { "skills_required": ["Java", "payment gateway"], "location": "Pune", "min_years": 3 }, "results": [{ "full_name": "Rahul Sharma", "match_score": 94, "similarity": 0.94, "explanation": "Expert Java dev (6 yrs)...", "top_skills": ["Java", "Spring Boot", "Razorpay" ] }, {"total": 12 }
```